Expected Value and Variance

Section 6.4

Video 79: Random Variables

- Random Variables
- Examples

Motivation

Often we are not interested in the outcomes of an experiment per se, but some function of those outcomes

The sum of the values of two dices

The money we win when rolling two dices summing up to 7

The number of 6 we roll in 10 rolls

Random Variables

Definition: A **random variable** X is a function X: S → **R** from the sample space S of an experiment to the set of real numbers **R**.

- A random variable assigns a real number to each possible outcome
- A random variable is a function.
- It is not a variable, and it is not random!
- In the late 1940s W. Feller and J.L. Doob flipped a coin to see whether both would use “random variable” or the more fitting “chance variable.” Unfortunately, Feller won and the term “random variable” has been used ever since.

Example

Suppose that a coin is flipped three times.

Let $X(t)$ be the random variable that equals the number of heads that appear when t is the outcome.

$$C = \{H, T\}, S = C^n, X : S \to R$$

Then $X(t)$ takes the following values:

$X(HHH) = 3$

$X(HHT) = X(HTH) = X(THH) = 2$

$X(TTH) = X(THT) = X(HTT) = 1$

$X(TTT) = 0$

Distribution of a Random Variable

Definition: The **distribution** of a random variable X on a sample space S is the set of pairs $(r, p(X = r))$ for all $r \in X(S)$, where $p(X = r)$ is the probability that X takes the value r

$$p(X = r) = \sum_{s \in S : X(s) = r} p(s)$$

- Assigns a probability to each possible value of the random variable
- $p(X = r)$ can be interpreted as a function $p : X(S) \to R$

Example

Suppose that a coin is flipped three times.
Let $X(t)$ be the random variable that equals the number of heads that appear when t is the outcome.

Each of the eight possible outcomes has probability 1/8.
So, the distribution of $X(t)$ is given by
$p(X = 3) = 1/8$, $p(X = 2) = 3/8$, $p(X = 1) = 3/8$, $p(X = 0) = 1/8$
and the distribution is
$\{(3, 1/8), (2, 3/8), (1, 3/8), (0, 1/8)\}$

Bernoulli Trial

The number of successes in *n* Bernoulli trials is

$$p(X = k) = C(n, k)p^k q^{n-k} = b(k : n, p)$$

Example: probability of rolling 3 times a 6 in 10 rolls

$$p(X = 3) = C(10,3)(1/6)^3(5/6)^7 \approx 0.155$$

Summary

- Random Variables
- Examples
  - Coin flips
  - Bernoulli trials

Video 80: Expected Value

- Expected Value
- Examples

Expected Value

Definition: The **expected value** (or **expectation** or **mean**) of the random variable *X* on the sample space *S* is equal to

$$E(X) = \sum_{s \in S} p(s)X(s)$$

Example

Expected Value of a Dice: Let *X* be the number that comes up when a fair dice is rolled. What is the expected value of *X*?

The random variable *X* takes the values 1, 2, 3, 4, 5, or 6.
Each has probability 1/6.
It follows that

$$E(X) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \ldots + \frac{1}{6} \cdot 6 = \frac{21}{6} = \frac{7}{2}$$

Expected Value

Theorem 1: If X is a random variable and $p(X = r)$ is the probability distribution with $p(X = r) = \sum_{s \in S, X(s)=r} p(s)$ then

$$E(X) = \sum_{r \in X(S)} p(X = r) r$$

Proof:

$$E(X) = \sum_{s \in S} p(s) X(s) \qquad \text{definition of expectation value}$$

$$= \sum_{r \in X(s)} \sum_{s \in S, X(s)=r} p(s) X(s) \qquad \text{regrouping summation}$$

$$= \sum_{r \in X(s)} \sum_{s \in S, X(s)=r} p(s) r \qquad X(s) \text{ constant in inner sum}$$

$$= \sum_{r \in X(s)} r \sum_{s \in S, X(s)=r} p(s) \qquad \text{factoring out } r$$

$$= \sum_{r \in X(s)} r p(X = r) \qquad \text{definition of } p(X = r)$$

Example

What is the expected value of the sum of the numbers that appear when a pair of fair dice is rolled?

Let X be the random variable equal to the sum of the numbers that appear when a pair of fair dice is rolled.

The range of X is {2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}

$$
\begin{aligned}
&p(X = 2) = p(X = 12) = 1/36,\\
&p(X = 3) = p(X = 11) = 2/36 = 1/18,\\
&p(X = 4) = p(X = 10) = 3/36 = 1/12,\\
&p(X = 5) = p(X = 9) = 4/36 = 1/9,\\
&p(X = 6) = p(X = 8) = 5/36,\\
&p(X = 7) = 6/36 = 1/6.
\end{aligned}
$$

therefore

$$
\begin{aligned}
E(X) = {} & 2 \cdot \frac{1}{36} + 3 \cdot \frac{1}{18} + 4 \cdot \frac{1}{12} + 5 \cdot \frac{1}{9} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{1}{6}\\
& + 8 \cdot \frac{5}{36} + 9 \cdot \frac{1}{9} + 10 \cdot \frac{1}{12} + 11 \cdot \frac{1}{18} + 12 \cdot \frac{1}{36}\\
= {} & 7.
\end{aligned}
$$

Expected Value of Bernoulli trials

Theorem 2: The expected number of successes when n mutually independent Bernoulli trials are performed, where p is the probability of each trial, is np.

Proof

Let X be the random variable equal to the number of successes in n trials. Then $p(X = k) = C(n, k)\, p^k q^{n-k}$. We have.

$$E(X) = \sum_{k=1}^{n} kp(X = k) \qquad \text{by theorem 1}$$

$$= \sum_{k=1}^{n} kC(n,k)p^k q^{n-k} \qquad kC(n,k) = k\frac{n(n-1)\ldots(n-k+1)}{k(k-1)\ldots 1} = nC(n-1,k-1)$$

$$= \sum_{k=1}^{n} nC(n-1,k-1)p^k q^{n-k}$$

$$= np\sum_{k=1}^{n} C(n-1,k-1)p^{k-1}q^{n-k}$$

$$= np\sum_{j=0}^{n-1} C(n-1,j)p^j q^{n-1-j} \qquad \text{replacing } k-1 \text{ by } j$$

$$= np(p+q)^{n-1} \qquad \text{binomial theorem}$$

$$= np \qquad p + q = 1$$

Summary

- Expected Value
- Examples
    - Expected Value of Bernoulli trials

Video 81: Linearity of Expectation

- Properties of Expected Value
- Independent Random Variables

Linearity of Expectations

Theorem 3: If X_i, $i = 1, 2, \ldots, n$ with n a positive integer, are random variables on S, and if a and b are real numbers, then

(*i*) $E(X_1 + X_2 + \ldots + X_n) = E(X_1) + E(X_2) + \ldots + E(X_n)$

(*ii*) $E(aX + b) = aE(X) + b$.

- The theorem tells that expected values are linear.

Proof

Proof of (i), for the case i = 2:

$$E(X_1 + X_2) = \sum_{s \in S} p(s)(X_1(s) + X_2(s)) = \sum_{s \in S} p(s)X_1(s) + \sum_{s \in S} p(s)X_2(s)) = E(X_1) + E(X_2)$$

The general case follows by mathematical induction.

Proof of (ii):

$$E(aX + b) = \sum_{s \in S} p(s)(aX(s) + b) = a \sum_{s \in S} p(s)X(s) + b \sum_{s \in S} p(s) = aE(X) + b$$

since $\sum_{s \in S} p(s) = 1$

Example

We have shown that the expected value of rolling a single dice is 7/2

Thus the expected value of the sum of rolling two dice is 7/2 + 7/2 = 7

This is a simpler way to compute the expected value for two dice

Inversions in a Permutation

Definition: The ordered pair (i, j) is an **inversion** in a permutation of the first n positive integers if $i < j$, but j precedes i in the permutation.

Example: There are six inversions in the permutation of 3, 5, 1, 4, 2

$$(1, 3), (1, 5), (2, 3), (2, 4), (2, 5), (4, 5).$$

Problem: Find the expected number of inversions in a random permutation of the first n integers.

Expected Number of Inversions

Let $I_{i,j}$ be the random variable on the set of all permutations of the first n positive integers with $I_{i,j} = 1$ if (i, j) is an inversion of the permutation and $I_{i,j} = 0$ otherwise.

Let X be the random variable equal to the number of inversions in the permutation:

$$X = \sum_{1 \le i < j \le n} I_{i,j}$$

It is equally likely for *i* to precede *j* in a randomly chosen permutation as it is for *j* to precede *i*, therefore for all (i, j)

$$E(I_{i,j}) = 1 \cdot p(I_{i,j} = 1) + 0 \cdot p(I_{i,j} = 0) = 1 \cdot 1/2 + 0 \cdot 1/2 = 1/2$$

There are $\binom{n}{2}$ pairs (i, j) with $1 \le i < j \le n$ and so we obtain the expected number of inversion using linearity of expectation

$$E(X) = \sum_{1 \le i < j \le n} E(I_{i,j}) = \sum_{1 \le i < j \le n} 1/2 = \binom{n}{2} 1/2 = \frac{n(n-1)}{4}$$

Independent Random Variables

Definition 3: The random variables *X* and *Y* on a sample space *S* are independent if

$$p(X = r_1 \wedge Y = r_2) = p(X = r_1) \cdot p(Y = r_2)$$

Theorem 5: If *X* and *Y* are independent variables on a sample space *S*, then

$$E(X \cdot Y) = E(X) \cdot E(Y)$$

In general, multiplicativity of expectation values does not hold!

Proof

$$E(XY) = \sum_{r \in XY(S)} rp(XY = r) = \sum_{r_1 \in X(S), r_2 \in Y(S)} r_1 r_2 p(X = r_1 \wedge Y = r_2) = \sum_{r_1 \in X(S)} \sum_{r_2 \in Y(S)} r_1 r_2 p(X = r_1 \wedge Y = r_2) =^*$$

$$= \sum_{r_1 \in X(S)} \sum_{r_2 \in Y(S)} r_1 r_2 p(X = r_1) p(Y = r_2) = \sum_{r_1 \in X(S)} (r_1 p(X = r_1) \sum_{r_2 \in Y(S)} r_2 p(Y = r_2)) = \sum_{r_1 \in X(S)} r_1 p(X = r_1) = E(Y) \sum_{r_1 \in X(S)} r_1 p(X = r_1) = E(Y)E(X)$$

* here we use the independence assumption

Summary

- Properties of Expected Value
  - Linearity
  - Inversions of Permutations
- Independent Random Variables
  - Multiplicativity

Video 82: Variance

- Variance
- Examples

Variance

Definition 4: Let X be a random variable on the sample space S. The **variance** of X, denoted by $V(X)$ is

$$V(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s)$$

The **standard deviation** of *X*, denoted by $\sigma(X)$ is defined as $\sqrt{V(X)}$

- Variance and standard deviation are used to quantify how widely a random variable is distributed

Example

Let X and Y be random variables on S = {1, 2, 3, 4, 5, 6}

Let X(s) = 0 for all $s \in S$
Let Y(s) = -1 for $s \in \{1,2,3\}$ and Y(s) = 1 for $s \in \{4,5,6\}$

Then E(X) = E(Y) = 0

But V(X) = 0 and V(Y) = 1

Characterisation of Variance

Theorem 6: If X is a random variable on a sample space S, then

$$V(X) = E(X^2) - E(X)^2$$

Corollary 1: If *X* is a random variable on a sample space *S* and *E*(*X*) = μ , then

$$V(X) = E((X - \mu)^2)$$

Proof of Theorem

$$
\begin{aligned}
V(X) &= \sum_{s \in S} (X(s) - E(X))^2 p(s) \\
&= \sum_{s \in S} X(s)^2 p(s) - 2E(X) \sum_{s \in S} X(s) p(s) + E(X)^2 \sum_{s \in S} p(s) \\
&= E(X^2) - 2E(X)E(X) + E(X)^2 \\
&= E(X^2) - E(X)^2.
\end{aligned}
$$

Proof of Corollary

$$E((X-\mu)^2) = E(X^2 - 2\mu X + \mu^2) =^* E(X^2) - E(2\mu X) + E(\mu^2) =^*$$

$$= E(X^2) - 2\mu E(X) + E(\mu^2) = ^{**} (X^2) - 2\mu E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2 = V(X)$$

* Linearity of expectation value

** μ^2 is a constant, therefore $E(\mu^2) = \mu^2$

Example

Variance of the Value of a Die: What is the variance of a random variable X, where X is the number that comes up when a fair die is rolled?

We have $V(X) = E(X^2) - E(X)^2$.

We have shown that $E(X) = 7/2$.

We calculate $E(X^2) = 1/6\ (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = 91/6$

and obtain $V(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$

Variance of Bernoulli Trials

What is the variance of the random variable X, where $X(t) = 1$ if a Bernoulli trial is a success and $X(t) = 0$ if it is a failure, where p is the probability of success and q is the probability of failure?

Because X takes only the values 0 and 1, it follows that $X^2(t) = X(t)$. Hence,

$$V(X) = E(X^2) - E(X)^2 = p - p^2 = p(1 - p) = pq.$$

Variance for Independent Random Variables

Bienaymé's Formula: If X and Y are two independent random variables on a sample space S, then $V(X + Y) = V(X) + V(Y)$.

Furthermore, if X_i, $i = 1,2, \ldots,n$, with n a positive integer, are pairwise independent random variables on S, then

$$V(X_1 + X_2 + \cdots + X_n) = V(X_1) + V(X_2) + \cdots + V(X_n).$$

Proof

$$
\begin{aligned}
V(X+Y) &= E((X+Y)^2) - E(X+Y)^2 = E(X^2 + 2XY + Y^2) - (E(X) + E(Y))^2 \\
&= E(X^2) + 2E(XY) + E(Y^2) - E(X)^2 - 2E(X)E(Y) - E(Y)^2 \\
&= V(X) + V(Y) + 2(E(XY) - E(X)E(Y))
\end{aligned}
$$

This shows that in case X and Y are independent that variance is additive.

Example

Find the variance of the number of successes when n independent Bernoulli trials are performed, where on each trial, p is the probability of success and q is the probability of failure.

Let X_i be the random variable with $X_i((t_1, t_2, \ldots, t_n)) = 1$ if trial t_i is a success and $X_i((t_1, t_2, \ldots, t_n)) = 0$ if it is a failure.

Let $X = X_1 + X_2 + \ldots X_n$.

Then X counts the number of successes in the n trials.

By Bienaymé 's Formula, it follows that $V(X) = V(X_1) + V(X_2) + \cdots + V(X_n)$.

We have shown that $V(X_i) = pq$ for $i = 1,2, \ldots, n$.

Hence, $V(X) = npq$.

Summary

- Variance
  - Definition
  - Characterisation using expected value
- Examples
  - Bernoulli trials
  - Independent random variables

Video 83: Estimating Deviations

- Markov’s Inequality
- Chebyshev’s Inequality
- Examples

Markov's Inequality

Let X be a non-zero and non-negative random variable:

$$\exists s X(s) > 0 \wedge \forall s X(s) \geq 0$$

Let $p(X \geq a)$ denote the probability that the variable attains a value larger than a

Then $\forall M > 0 \; p(X \geq M \cdot E(X)) \leq \dfrac{1}{M}$

Proof

Proof: denote $a = M \cdot E(X)$

$$\frac{1}{M} = \frac{E(X)}{a} = \sum_{s \in X(S)} xp(X = x)/a$$

$$\geq \sum_{s \in X(S), x \geq a} xp(X = x)/a \quad X \text{ is non-negative}$$

$$\geq \sum_{s \in X(S), x \geq a} p(X = x) \quad \text{since } x \geq a \text{ implies } x/a \geq 1$$

$$= p(X \geq a)$$

Example

Rolling a dice, where rolling a 6 is considered as success (Bernoulli trial)

When performing 6 trials the expectation value is to obtain 1 success.

Estimate the probability of having at least 3 successes:

$$p(X \geq 3) = p(X \geq 3E(X)) \leq \frac{1}{3}$$

Chebyshev's Inequality

Let X be a random variable on a sample space S with probability function p. If r is a positive real number, then

$$p(|X(s) - E(X)| \geq r) \leq V(X)/r^2$$

Proof

Let A be the event $A = \{s \in S \mid |X(s) - E(X)| \geq r\}$

Note that

$$V(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s) = \sum_{s \in A} (X(s) - E(X))^2 p(s) + \sum_{s \notin A} (X(s) - E(X))^2 p(s)$$

$$\geq \sum_{s \in A} (X(s) - E(X))^2 p(s) \geq \sum_{s \in A} r^2 p(s) = r^2 p(A)$$

Therefore $p(A) = p(|X(s) - E(X)| \geq r) \leq V(X)/r^2$

Example

Rolling a dice, where rolling a 6 is considered as success (Bernoulli trial)

When performing 6 trials the expectation value is to obtain 1 success.
The variance is *npq* = 6*1/6*5/6 = 5/6

Estimate the probability of having at least 3 successes:

$$p(|X(s) - E(X)| \geq 2) \leq V(X)/2^2 = \frac{5}{6 \cdot 2^2} = \frac{5}{24} \approx \frac{1}{5}$$

Note: this is a smaller bound than ⅓ obtained with the Markov inequality

Example

We can also compute the exact probability of having 3 or more successes using the binomial distribution

$$b(3:6,1/6)+b(4:6,1/6)+b(5:6,1/6)=\frac{6\cdot 5\cdot 4}{1\cdot 2\cdot 3}\frac{5^3}{6^6}+\frac{6\cdot 5}{1\cdot 2}\frac{5^2}{6^6}+\frac{6}{1}\frac{5}{6^6}\approx 0.06$$

Thus also the estimate using Chebyshev's Inequality was not very sharp

Example

Suppose that X is a random variable that counts the number of tails when a fair coin is tossed n times.

Note that X is the number of successes when n independent Bernoulli trials, each with probability of success ½ are done.

Hence, $E(X) = np = n/2$ and $V(X) = npq = n/4$.

By Chebyshev's inequality with $r = \sqrt{n}$,

$$p(|X(s) - n/2| \geq \sqrt{n}) \leq (n/4)(\sqrt{n})^2 = ¼.$$

This means that the probability that the number of tails that come up on n tosses deviates from the mean of $n/2$ by more than $\sqrt{n}$, is no larger than ¼.

Summary

- Markov’s Inequality
  - based on expectation value
- Chebyshev’s Inequality
  - based on variance